

Financial Functions, Logical Functions.

① How do you calculate loan payments using the PMT functions?

→ The PMT function in excel is used to calculate the periodic payment required to repay a loan based on a constant interest rate and fixed number of periods. The syntax is

$\text{PMT}(\text{rate}, \text{nper}, \text{PV}, [\text{FV}], [\text{type}])$.

Here, rate is the interest rate per period, nper is the total number of payments, and PV is the loan amount.

for example, to calculate the monthly payment on a
£ 500000 loan at 10% annual interest for 5 years,
the formula is
$$= \text{PMT}(10\%/12, 5*12, 500000)$$

what is the difference between the NPV and
IRR functions?

The NPV (net present value) function calculates the
present value of future cash flows using a given
discount rate, helping to decide whether an investment
is profitable. A positive NPV indicates investment
is

The IRR (Internal Rate of Return) function calculates
the discount rate at which the NPV of cash
flows becomes zero. It represents the expected
rates of return of an investment.
while NPV gives a value, IRR gives a
percentage.

③ Explain how the FV function is used to
calculate the future value of an investment.

The FV (future value) function is used to
calculate the value of an investment at a future
date based on constant investment at a future
interest rate and periodic payments.

Syntax: $= \text{FV}(\text{rate}, \text{nper}, \text{Pmt}, [\text{pv}], [\text{type}])$

It is commonly used for savings plans, investments and loan analysis, for example, It helps calculate how much money will be accumulated after paying a fixed amount regularly for a specific period.

④ How would you use the IF function to perform Conditional Calculations?

→ The IF function performs logical tests and returns different results based on whether the condition is TRUE or FALSE

Syntax: =IF (logical-Test, value-if-true, value-if-false)

It is used for decision-making tasks such as grading students, determining pass or fail status, or applying discounts based on conditions.

⑤ What are nested IF Statements, and how can they be applied?

→ Nested IF statements are multiple IF functions placed inside one another to test more than one condition.

They are used when there are multiple possible outcomes, for example, assigning grades such as A, B, C or fail based on marks.

Nested IF allows complex logical decision-making within a single formula.

⑥ How do you the AND and OR functions in combination with IF?

- The And and or functions are used with IF to test multiple conditions
- AND returns TRUE if all conditions are met
 - OR returns TRUE if at least one condition is met.

Example: $=IF(AND(A1 \geq 40, B1 \geq 40), "pass", "Fail")$

← This combination helps in advanced logical analysis.

⑦ What is the purpose of the IFERROR function, and how does it work?

- The IFERROR function is used to handle error in formulas and display a custom message instead of an error value.

Syntax: $=IFERROR(\text{value}, \text{value if error})$

It replaces error like #DIV/0!, #NA, or

#value with meaningful text such as "invalid" or "0" making spreadsheets cleaner and easier to understand.

⑧ Explain the difference between the ISNUMBER and ISTEXT functions.

→ The ISNUMBER function checks whether a value is numeric and returns TRUE if it is a number.

The ISTEXT function checks whether a value is text and returns TRUE if it contains text.

These functions are useful for data validation, cleaning data and applying formulas only to specific data types.

⑨ How would you use the CUMIPMT function in Excel?

→ The CUMIPMT function calculates the cumulative interest paid on a loan between two periods.

Syntax: ~~rate~~ = CUMIPMT (rate, nper, PV, start period, end - period, type)

It is mainly used in financial analysis to find total interest paid over a specific time range of a loan or mortgage.

⑩ Can you describe a scenario where the SLN function would be useful?

→ The SLN (Straight - line Depreciation) function is used to calculate depreciation of an asset evenly over its useful life.

It is useful in accounting for assets like machinery, vehicles, or equipment.

For example, a company can use SLN to calculate yearly depreciation expense for a machine over 10 years.